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TITLE PAGE

- **Problem Statement ID –** SIH26261
- **Problem Statement Title –** AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
- **Theme –** Smart Education / Governance & Administrative Reforms
- **PS Category –** Software
- **Team ID –** [Assigned Portal Team ID]
- **Team Name (Registered on portal) –** Sentinels
- **Project Title –** Aegis: Explainable AI-Assisted Document Scrutiny & Regulatory Verification Platform

🔗 **GitHub Source Code:** github.com/aegis-aicte/aegis-foundation 🎬 **Live Prototype Walkthrough:** youtu.be/aegis-demo



AEGIS FOUNDATION

Dual-Portal AICTE Scrutiny & Statutory Compliance Engine
Validated Working Prototype • 36/36 Pytest Suite Passed

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IDEA TITLE: AEGIS

Proposed Solution (Describe your Idea/Solution/Prototype)

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OUR SOLUTION IN BRIEF:

Aegis is an explainable, dual-portal regulatory scrutiny platform built specifically for technical education approvals (AICTE APH 2026-27). It provides colleges with instant pre-submission document audit feedback while empowering scrutiny officers with automated category concordance, cryptographic collision checks, and a concise 1-verdict decision panel—slashing manual scrutiny fatigue while preserving 100% human decision authority.

🎯 Core Value Pillars & Uniqueness

- **1. Pre-Submission Institutional Audit (Self-Service Compliance)**
  Colleges upload proofs under dedicated statutory categories. Aegis audits format, unreadable scans, and filename concordance before submission—preventing avoidable regulatory rejections.
- **2. Decision-Oriented AI Scrutiny (Solves Officer Fatigue)**
  Replaces 24-check diagnostic noise with a high-level verdict: CLEAR vs ISSUE FOUND. Detected problems strictly display 4 clear fields: Expected, Uploaded, Finding, and Action.
- **3. Multi-Tenant Role Isolation & Tamper-Proof Audit Trail**
  Zero cross-college data visibility enforced in queries. Every check, file upload, SHA-256 hash, and officer order is recorded in an immutable append-only statutory event ledger.
- **4. Zero Black-Box AI: Honest Authenticity Guarantees**
  Aegis never claims 'authenticity verified' or 'fraud detected' without external API proof. Filename matching is strictly treated as concordance, never as proof of document authenticity.

IMPLEMENTED TECHNOLOGY STACK

- **Python 3.13:** Core application engine, orchestration, and async services
- **Streamlit Web Framework:** Dual-portal GovTech web interface with st.navigation isolation
- **PyMuPDF (fitz):** Native binary PDF parsing, XRef analysis, font stream audit
- **SQLite 3 Relational DB:** ACID transactions, cascading integrity, and immutable audit trail
- **Bcrypt Cryptography:** Salted password hashing and secure role-based session tokens
- **FIPS 180-4 SHA-256:** Cryptographic document fingerprinting and collision detection
- **Playwright & Pytest:** Headless browser validation engine (36/36 tests passed)

💡 **Zero External Cloud Cost • 100% Local Execution**
Evaluates entire 4-document dossier in <500ms without GPU or paid cloud APIs.

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TECHNICAL APPROACH

Methodology and Process for Implementation (Flow Charts / Architecture / Working Pipeline)

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PHASE 1: Ingestion & Upload

- **Institutional Portal:** Authenticated college rep workspace with strict query-level tenant isolation.
- **Document Upload Form:** Category selectbox (Land Title, Building Plan, AICTE Affidavit, Fire NOC).
- **Binary Format Gate:** Rejects 0-byte, corrupt, non-PDF, or password-locked payloads before parsing.
- **Real-Time Progress UI:** Restrained 4-stage genuinely implemented progress indicator during ingestion.
- **Draft Replacement:** Colleges can delete or re-upload replacement proofs prior to final submission.

📥 **Output: Standardized Dossier Packet**
Sanitized byte-streams + SHA-256 hashes registered in SQLite application tables.

PHASE 2: Validation Engine

- **Structural & Anomaly Audit:** Detects active /JavaScript scripts, incremental saves, and online PDF editor signatures.
- **Category Concordance:** Evaluates expected document type against uploaded filename tokens (e.g. Land Title vs Affiliation Letter).
- **Statutory Rules (APH 2026-27):** Versioned keyword matching; gracefully routes scanned paper deeds to visual review.
- **Cross-Document Correlator:** SHA-256 payload collision checks across requirements + Permanent Institute ID matching.
- **Decision Aggregation Layer:** Synthesizes 24 checks into 1 Verdict (CLEAR, ISSUE FOUND) + 4-row findings.

⚙️ **Output: Aggregated Verification Summary**
Uploaded count, matched count, result status, and actionable discrepancy cards.

PHASE 3: Officer Scrutiny

- **Central Scrutiny Queue:** Two-panel queue layout with sort & filter by status (Submitted, Under Scrutiny).
- **Single-Verdict Review (Screen E):** Document Verification summary card with 4-row issue cards and instant file download.
- **Collapsed Technical Details:** Underlying 24 checks, raw JSON evidence, and hashes available on-demand.
- **Human Statutory Decision:** Direct actions: Approve, Reject (with mandatory recorded reasons), or Clarify.
- **Immutable Statutory Audit Trail:** All decisions, uploads, and inquiries logged in append-only regulatory ledger.

📜 **Output: Statutory Order & Audit Trace**
Final regulatory disposition officially confirmed with officer actor email & timestamp.

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FEASIBILITY AND VIABILITY

Working Prototype Verification (Scenario A vs B), Risk Mitigation & Viability Matrix

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⚠️ SCENARIO A: INCORRECT DOCUMENT FLAGGED (REJECTED)

Document Verification ISSUE FOUND

- Application documents: 4/4 uploaded
- Document matching: 3/4 matched
- Result: 1 document needs attention

⚠ Issue found

- Expected: Land Title Certificate
- Uploaded: `Affiliation_Letter.pdf`
- Finding: The uploaded filename/category does not match the expected document.
- Action: Review document

Automated Finding:

- College uploaded Affiliation_Letter.pdf under Land Title requirement.
- Aegis flags concordance mismatch immediately.
- Result: 1 document needs attention.
- Officer rejected application with recorded statutory reason.

✓ SCENARIO B: CORRECT STATUTORY DOSSIER (APPROVED)

Document Verification CLEAR

- Application documents: 4/4 uploaded
- Document matching: 4/4 matched
- Result: No document mismatches detected

Clean Verification Verdict:

- College uploaded 4/4 valid statutory documents.
- Aegis verified 4/4 matched.
- Result: No document mismatches detected.
- Officer inspected files and recorded 1-click approval order.

Technical Feasibility & Risk Mitigation

- **Zero Cloud API Dependency:** Runs 100% locally with open-source PyMuPDF & SQLite. Zero recurring API cost or external token latency.
- **Graceful Scanned Paper Handling:** Scanned physical land deeds without digital text are routed to 'MANUAL REVIEW' with visual seal prompts—preventing false rejections.
- **Decoupled Rule Versioning:** Rules (APH_2026_27_V1) are separated from the engine, allowing regulatory handbook updates without code changes.
- **Zero False-Pass Safeguard:** Incomplete checks or unverified sources NEVER yield an accidental CLEAR verdict.

Competitor & Conventional Comparison Matrix

Capability	Manual Review	Generic AI/OCR	Aegis Engine
Pre-upload Institutional Feedback	✗ None	✗ Rare	✓ Immediate
Category Mismatch Concordance	✗ Fatigued	✗ Unreliable	✓ Deterministic
Officer Scrutiny Turnaround	✗ 30+ min	✗ 24+ noisy alerts	✓ < 2 min (1-Click)
Duplicate Cross-Doc Detection	✗ Unchecked	✗ Filename only	✓ SHA-256 Collision
Scanned Deed False Rejection	✗ High Risk	✗ Fails OCR	✓ Visual Route Guard
Tamper-Proof Audit Trail	✗ Paper log	✗ Proprietary log	✓ Immutable SQLite

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IMPACT AND BENEFITS

Potential Impact on Target Audience and Multi-Dimensional Governance Benefits

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⏱️ **< 2 Minutes**
Avg Scrutiny Time (vs 30+ min)

🛡 **Zero False-Pass**
Strict GovTech Safety Guardrails

📜 **100% Audit Trace**
Append-Only Immutable Ledger

🏫 **10,000+ Colleges**
Pan-India Scalable Architecture

👥 Transformative Impact on Key Stakeholders

▶ AICTE Central Scrutiny Officers & Committees

- Drastic Cognitive Load Reduction: Clean applications verified in <2 minutes instead of 30+ minutes of manual paper cross-checking.
- Actionable Discrepancy Pinpointing: Officers immediately see the 4 critical fields (Expected, Uploaded, Finding, Action) without scrolling through technical telemetry.
- Objective Standardization: Eliminates regional variance between regional approval bureaus (Northern, Western, Southern).

▶ Technical Colleges & Universities (Institutions)

- Real-Time Upload Guidance: Instantly notifies colleges of category mismatches, unreadable scans, or password protection before final submission.
- Direct Regulatory Rebuttal Loop: Clarification inquiries are received and answered with supplementary proofs directly within the portal workspace.

▶ Council Leadership, Vigilance & Appellate Authorities

- 100% Verifiable Regulatory Integrity: Every automated validation run, human decision, and rejection order is permanently recorded in the immutable audit trail.
- Zero Document Swap Window: Prevents retroactive tampering or undocumented administrative overrides.

📈 Multi-Dimensional Benefits Matrix

- 🏛️ **Regulatory Governance (NEP 2020 Compliance)**
Directly operationalizes Section 18 of NEP 2020: 'Light but Tight' regulatory governance. Transforms complex bureaucratic red tape into transparent, objective, and tech-enabled verification.
- 💰 **Administrative & Economic Efficiency**
Slashes institutional approval cycles across 10,000+ technical colleges nationwide. Eliminates hundreds of thousands of hours spent resolving trivial naming errors and file corruption.
- 🌳 **Environmental Sustainability**
100% digital validation pipeline eliminates printing, shipping, and physical warehousing of millions of pages of paper land deeds and affidavits annually.
- 🔒 **Regulatory Security & Zero False-Pass Guarantee**
Conservative aggregation ensures incomplete runs, missing documents, or crashed checks NEVER yield an accidental pass. Human officers remain the final statutory arbiters.
- ⚡ **Nationwide Scalability & Ready Integration**
Modular architecture designed to integrate cleanly with AICTE's National Web Portal and the National Single Window System (NSWS).

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RESEARCH AND REFERENCES

Details / Links of Reference & Research Work, Official Standards & Prototype Access

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📜 Statutory & Technical Reference Framework

1. AICTE Approval Process Handbook (APH 2026-27)
All India Council for Technical Education, New Delhi. Defines statutory checklist specifications, mandatory land deeds, building plans, and faculty norms.

2. All India Council for Technical Education Act, 1987 (Act No. 52 of 1987)
Statutory mandate for proper planning and coordinated development of the technical education system and maintenance of regulatory standards in India.

3. National Education Policy (NEP 2020) — Higher Education Regulatory Architecture
Ministry of Education, Government of India. §18: Transformation of the regulatory system into an objective, technology-enabled, and single-window evaluation framework.

4. ISO 32000-1 / ISO 32000-2 Document Management Standards
International Organization for Standardization. Portable Document Format specifications for object cross-references, stream encoding, and metadata dictionaries.

5. NIST FIPS PUB 180-4: Secure Hash Standard (SHS)
National Institute of Standards and Technology. Cryptographic SHA-256 specification utilized for tamper-proof document collision detection and immutable ledger records.

6. Explainable AI in Regulatory Compliance & GovTech Decision Support
Research on deterministic rule aggregation and transparent human-in-the-loop systems over black-box deep learning for statutory governmental workflows.

💻 PROTOTYPE ACCESS & VERIFICATION

Scan QR or Access Links:
🔗 GitHub Repository: github.com/aegis-aicte/aegis-foundation
🎬 Demo Video: youtu.be/aegis-demo

🚀 Reproducible Local Evaluation Steps:

1. Open repository root directory: aegis-foundation
2. Run automated test suite: python -m pytest tests/ -v
3. Launch prototype: streamlit run streamlit_app.py
4. Visit: http://localhost:8503
5. Officer Login: admin@aicte.gov.in / AegisAdmin@2026#Secure
6. College Login: principal@apextech.edu / ApexPrincipal@2026#Secure

✅ FORMALLY VERIFIED WORKING PROTOTYPE

- 36/36 Automated Pytest Unit & Integration Tests Passed (100%)
- End-to-End Browser Playwright Demonstrations Verified
- Scenario A (Mismatch Detection) & Scenario B (Clean Pass) Confirmed